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Consumer resistance and behavioral responses to front-of-pack nutrition labels: a systematic review through cognitive dissonance and psychological reactance frameworks

Carolina S. Portel [a,*], Júlia B. Sousa [b], Carolina P.C. Martins [b], Mônica M. Pagani [c], Eliane T. Mársico [b], Mônica Q. de Freitas [b], Adriano G. da Cruz [a], Erick A. Esmerino [a,b,**]

[a] Department of Food, Federal Institute of Education, Science and Technology of Rio de Janeiro (IFRJ), 20270-021, Rio de Janeiro, Rio de Janeiro, Brazil
[b] Department of Food Technology, Federal Fluminense University, Faculty of Veterinary, 24230-340, Niterói, Rio de Janeiro, Brazil
[c] Department of Food Technology, Federal Rural University of Rio de Janeiro, Faculty of Food Engineering, 23897-001, Seropédica, Rio de Janeiro, Brazil

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ABSTRACT

Background: Front-of-pack nutrition labeling (FOPNL) has been increasingly adopted across several countries as a public health policy aimed at improving dietary choices and reducing the burden of diet-related diseases. By presenting simplified nutritional information at the point of purchase, FOPNL are expected to support informed consumer decisions and stimulate product reformulation by the food industry. However, empirical evidence regarding their effectiveness remains heterogeneous. While some studies report improvements in food choices, others identify limited effects or even defensive consumer responses.

Scope and approach: This systematic review synthesizes empirical evidence on consumer responses to front-of-pack labels through the theoretical lenses of cognitive dissonance and psychological reactance. Following PRISMA 2020 guidelines, studies were identified through systematic searches in *PubMed*, *Scopus*, *ScienceDirect*, and *Web of Science*. Eligible studies included experimental, quasi-experimental, qualitative, and mixed-methods research examining psychological responses to nutrition labeling interventions in food choice contexts.

Key findings and conclusions: Psychological reactance emerged when labels were perceived as threatening consumer autonomy, particularly in warning-based systems. Cognitive dissonance occurred when label information conflicted with prior beliefs about product healthfulness. Responses varied across food categories and were influenced by message framing, label format, and perceived costs of behavioral change. Defensive reactions did not necessarily prevent behavioral change, suggesting that resistance and increased risk awareness may coexist. By integrating insights from consumer psychology, this review provides insights into factors that may contribute to heterogeneous consumer responses to front-of-pack labeling (FOPL) policies sometimes succeed or fail and provides guidance for designing labeling strategies that reduce defensive responses while strengthening public health interventions.

1. Introduction

The increasing prevalence of overweight and obesity represents one of the most pressing global public health challenges. A recent meta-analysis including data from 73 countries estimated that approximately 37% of the global adult population is overweight, with prevalence reaching nearly 60% in high-income countries (Islam et al., 2024). Among the multiple factors contributing to this scenario, the growing consumption of ultra-processed foods has been consistently highlighted as a key driver, due to their widespread availability, high energy density, and unfavorable nutritional composition (Monda et al., 2024).

In response to these challenges, governments and international health organizations have increasingly adopted policy guidelines aimed at improving the food environment and supporting healthier food choices. Within this context, front-of-pack labeling (FOPL) has emerged as one of the most widely innovative regulatory strategies. Currently,

* Corresponding author.
** Corresponding author. Department of Food, Federal Institute of Education, Science and Technology of Rio de Janeiro (IFRJ), 20270-021, Rio de Janeiro, Rio de Janeiro, Brazil.
E-mail addresses: carol.portel21@gmail.com (C.S. Portel), eaesmerino@id.uff.br (E.A. Esmerino).

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there is no single, mandatory international standard for FOPL regarding format, content, and criteria used, with different systems adopted in various countries in Europe and the Americas, reflecting a growing international effort to make nutritional information more accessible at the point of sale (Kanter et al., 2018). The literature organizes these labels primarily by level of interpretation (interpretative and non-interpretative) and by type of information (nutrient-specific and summary). Interpretative labels translate the content into judgments or risk/health signals (such as warnings, traffic light colors, stars), while non-interpretative labels only present numbers and factual information. Regarding the type of information, summary labels provide an overall assessment of the food, usually through a calculation based on algorithms (e.g., a single nutritional quality seal). On the other hand, specific nutrient labels separately highlight the content of individual nutrients (such as sugar, fat, sodium) on the front panel (Batista et al., 2023). Table 1 presents a summary of the main front-of-pack labels used worldwide and their classifications.

By presenting simplified nutritional information directly on the front of food packages, these labels aim to facilitate informed decision-making and encourage healthier product reformulation by the food industry (Braesco & Drewnowski, 2023; Ganderats-Fuentes & Morgan, 2023). Evidence suggests that FOPL may influence consumer behaviour through multiple pathways, including increased salience of nutritional information and greater awareness of diet-related health risks (Cecchini & Warin, 2016; Grunert & Wills, 2007; Ikonen et al., 2020; Song et al., 2021).

Despite its growing adoption as a public health policy, empirical evidence regarding the effectiveness of nutrition labeling remains heterogeneous. While some studies report improvements in consumer choices and dietary quality, others identify limited behavioral changes or even unintended responses (Cecchini & Warin, 2016; Grunert & Wills, 2007; Ikonen et al., 2020; Song et al., 2021).

Table 1
Typology of the main front-of-pack nutrition labeling systems according to the type of information, interpretive format and information format.

FOPL System (Main Examples)	Countries/ Regions	Type of information	Interpretive level	Information format
Nutri-Score	France and other European countries	Summary indicator	Interpretive	Aggregated nutritional score (A–E)
Health Star Rating (HSR)	Australia and New Zealand	Hybrid	Hybrid	Aggregated star-rating score
Traffic light label	United Kingdom, Ecuador	Hybrid	Interpretive	Color-coded nutrient levels
Warning labels	Chile, Mexico, Brazil, Israel	Nutrient-specific	Interpretive	Warning symbol indicating excess nutrients
NutrInform Battery	England	Nutrient-specific	Interpretive	Battery icons showing percentage of daily reference intake
Guideline Daily Amounts/ Reference Intakes (GDA/RI)	Europe	Nutrient-specific	Non-interpretive	Numerical nutrient and energy values
Keyhole/ Healthy Choice logos e outros “healthy choice”	Nordic countries	Summary indicator	Interpretive	Positive endorsement symbol identifying healthier options within a category

Recent evidence suggests that consumer responses to front-of-pack nutrition labels are influenced not only by the information conveyed but also by how that information aligns with existing beliefs and other trusted sources. Directive and summary labels may generate conflicting nutrition information when their evaluations contradict consumers' prior knowledge or scientific evidence, reducing the perceived congruence and credibility of the information provided and decreasing consumers' willingness to adopt such labels. Particularly in the case of directive or algorithm-based summary labels, conflicting nutrition information may emerge when the evaluation provided by the label contradicts consumers’ prior beliefs, product-related expectations, or other trusted nutrition information sources. Under such circumstances, consumers may perceive the label as incongruent with their existing knowledge, which can undermine its credibility and reduce willingness to rely on it during food choice decisions (Mazzù et al., 2024).

Furthermore, consumers tend to express lower trust in algorithm-based evaluations, particularly when simplified nutritional judgments are perceived as inconsistent with individual goals or existing beliefs about food products (Mazzù et al., 2022). Likewise, when food-related communications are perceived as having a manipulative intent, consumers may become more skeptical of the information source and develop less favorable responses toward it (He et al., 2024). Together, these findings suggest that front-of-pack labels may evoke psychological processes that extend beyond information processing alone, highlighting the relevance of mechanisms such as cognitive dissonance and psychological reactance in shaping consumer responses.

Insights from social psychology provide useful theoretical frameworks for understanding these responses. Cognitive dissonance theory proposes that exposure to information that conflicts with existing beliefs, values, or behaviors generates psychological discomfort, motivating individuals to restore cognitive consistency either through behavioral change or through reinterpretation of the information (Festinger, 1957). In the context of food, this dissonance can occur, for example, when an individual values a healthy diet but consumes foods that they consider unhealthy, generating a conflict between knowledge, attitudes, and dietary practices (Ong et al., 2017) Empirical studies suggest that labels highlighting health, environmental, or ethical consequences of food consumption may activate such processes, potentially leading to behavioral adjustment when individuals resolve the dissonance through changes in their choices (Edenbrandt et al., 2021; Davies & Stanley, 2024; Bouwman et al., 2022).

Informational interventions may also trigger defensive responses. Psychological reactance theory posits that when individuals perceive persuasive messages as threatening their freedom of choice, they may experience motivational states aimed at restoring autonomy, often resulting in message rejection or behavioral resistance. In the context of nutrition labeling, messages perceived as prescriptive, moralizing, or controlling may therefore provoke defensive reactions that reduce the effectiveness of the intervention (Graça et al., 2025; Hughner & Dumitrescu, 2024). However, some evidence suggests a more complex dynamic, where the net impact on behavioral change may remain positive despite emotional resistance (Clarke et al., 2020; Malik & Ong, 2024).

In short, while cognitive dissonance arises from an internal conflict related to an individual's own beliefs, values, and behaviors, psychological reactance reflects a defensive response to perceived external threats to personal autonomy (Rains, 2013). Although conceptually distinct, these mechanisms may interact in the context of front-of-pack labeling, as consumers can simultaneously experience conflict with nutritional information while perceiving labeling messages as restrictive or controlling. Considering both perspectives is therefore important for understanding the diversity of consumer responses to nutrition labels, ranging from behavioral change and risk awareness to resistance, rationalization, and message rejection.

Although research on nutrition labeling has expanded considerably in recent years, existing studies remain fragmented across disciplinary

fields, food categories, and methodological approaches. Moreover, previous reviews have primarily focused on behavioral or nutritional outcomes, with limited attention to the psychological mechanisms that may explain why labeling interventions sometimes succeed and sometimes fail. Table 2 summarizes the main literature reviews on nutrition labeling published in recent years, highlighting their scope, outcomes assessed, and the limited consideration of psychological mechanisms such as cognitive dissonance and psychological reactance. In particular, there is a lack of systematic syntheses integrating evidence on cognitive dissonance and psychological reactance in the context of FOPL.

While previous reviews have provided important evidence regarding the effectiveness, consumer understanding, purchasing behavior, and policy implications of FOPL systems, none has systematically synthesized the literature through the combined theoretical lenses of cognitive dissonance and psychological reactance. This distinction defines the specific contribution of the present review and addresses an important gap in understanding the psychological mechanisms underlying heterogeneous consumer responses to front-of-pack nutrition labels.

Addressing this gap is important for advancing theoretical understanding of consumer responses to nutrition information and for improving the design of labeling policies. By examining how different labeling formats interact with psychological processes and food-related beliefs, it becomes possible to identify the conditions under which labels promote behavioral change or trigger resistance.

Therefore, this review aims to synthesize empirical evidence on how cognitive dissonance and psychological reactance have been used to explain consumer responses to FOPL across different food categories. Specifically, this review seeks to: (i) identify how these psychological mechanisms have been operationalized and investigated in the literature; (ii) examine the contexts and food categories in which nutrition labeling is associated with behavioral change or resistance; and (iii) identify research gaps and directions for future studies. To achieve these objectives, a systematic review of the literature was conducted following PRISMA 2020 guidelines, as described in the next section.

2. Methods

2.1. Study design

This study was conducted as a systematic review aimed at synthesizing empirical evidence on consumer responses to nutrition labeling through the theoretical lenses of cognitive dissonance and psychological reactance. The review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) guidelines (Page et al., 2021), which provide a standardized framework for transparent reporting of systematic reviews.

The methodological procedures included systematic database searches, predefined eligibility criteria, independent screening of studies, structured data extraction, and narrative synthesis of the findings. No formal review protocol was prospectively registered for this systematic review.

2.2. Search strategy

A systematic search of the literature was conducted in four electronic databases: PubMed, Scopus, Web of Science and ScienceDirect. These databases were selected due to their extensive coverage of research in nutrition, public health, consumer behavior, and food policy. A time restriction was applied to studies published from 2016 onwards, a period that coincides with the implementation and investigation of front-of-pack nutrition labeling systems in the scientific literature. The last search update was conducted on May 26, 2026.

The search strategy combined three main groups of keywords related to nutrition labeling, psychological mechanisms, and consumer food-related behaviors (Supplementary Table S1). Boolean operators were used to combine these groups of terms. In addition to database searches, complementary identification strategies were applied, including manual screening of reference lists of eligible articles and citation tracking to identify additional relevant studies. No language restrictions were applied. All records identified as potentially eligible at the full-text stage were available in English.

Table 2
Comparative overview of systematic reviews on FOPL.

Previous Review	Focus	Main findings	Limitations	Relevance to the present review
Croker et al. (2020)	Dietary purchasing and consumption behavior.	FOPL promoted healthier purchases, with lower sugar and sodium content and trends toward lower energy and saturated fat intake.	Did not explore psychological mechanisms underlying consumer.	Examines cognitive dissonance and psychological reactance as mechanisms influencing responses to FOPL.
Mazzù et al. (2022)	Mapping the FOPL literature through bibliometric and qualitative analyses.	Summary labels (like Nutri-Score) have improved objective understanding, while nutrient-specific labels have increased transparency and trust. Too much information negates the effectiveness of labels.	Focused on bibliometric trends and label performance, without addressing underlying psychological processes.	Investigates the psychological mechanisms that may explain why consumers accept, ignore, or resist FOPL.
Shrestha et al. (2023)	Socioeconomic differences in understanding and use of FOPL.	Individuals with higher socioeconomic status demonstrated better understanding and use of FOPL.	Did not examine psychological factors that may contribute to differential responses across consumer groups.	Highlights the potential relevance of psychological mechanisms for understanding heterogeneous consumer responses to FOPL.
Kelly et al. (2024)	Consumer understanding, food choices, purchasing behavior, and food reformulation.	FOPL improved understanding and healthier choices, with interpretive labels showing the strongest effects.	Focused on behavioral outcomes and effectiveness of label formats, without investigating psychological resistance mechanisms.	Investigates psychological processes that may explain both positive and negative responses to FOPL.
Guo et al. (2024)	Effectiveness of different FOPL formats among young adults.	Nutri-Score showed the strongest performance; warning labels produced mixed results; numeric labels were the least effective.	Did not assess motivational conflict, defensive reactions, or resistance to nutrition messages.	Synthesizes evidence on psychological mechanisms associated with responses to different FOPL formats.
Ge et al. (2026)	Effects of FOPL on social inequalities in sugar consumption and purchasing.	FOPL reduced sugar consumption and purchasing at the population level, but evidence regarding social inequalities remained inconclusive.	Did not investigate the psychological mechanisms that may contribute to unequal responses across population groups.	Examines cognitive and motivational processes that may help explain variability in consumer responses to FOPL.

Legend: FOPL = Front-of-Pack Labeling.

2.3. Eligibility criteria

Studies were considered eligible if they met predefined inclusion criteria related to study design, topic, psychological constructs, and study population. Only empirical studies employing experimental, quasi-experimental, qualitative, or mixed-methods designs were included, as these approaches allow for the direct investigation of consumer responses to nutrition labeling interventions. The studies needed to examine nutrition labeling applied to food or beverage products and to assess psychological responses related to cognitive dissonance or psychological reactance. Constructs associated with cognitive dissonance included psychological discomfort, guilt, rationalization, attitude change, or information avoidance, while indicators of psychological reactance included perceived threats to freedom of choice, emotional reactions such as anger or irritation, message rejection, or defensive responses to labeling information. The population of interest consisted of adult consumers aged 18 years or older exposed to nutrition labeling in real or simulated food choice contexts.

Studies were excluded if they consisted of narrative or systematic reviews, editorials, letters to the editor, or conference abstracts. Studies conducted in non-food contexts, such as tobacco or alcohol labeling, were also excluded. Additionally, studies that examined nutrition labeling effects without addressing psychological resistance mechanisms or those for which the full text was not available were not considered eligible for inclusion.

2.4. Study selection

The study selection process followed a two-stage screening procedure. Initially, duplicate records identified across databases were removed. Subsequently, titles and abstracts of the remaining records were screened to identify potentially relevant studies according to the predefined eligibility criteria. In the second stage, the full texts of all potentially eligible articles were retrieved and evaluated to determine their inclusion in the review.

The screening process was conducted independently by two reviewers in order to reduce the risk of selection bias. Any discrepancies between reviewers were resolved through discussion and consensus. When necessary, studies were re-evaluated jointly to ensure consistent application of the eligibility criteria. The complete study selection procedure is summarized using a PRISMA flow diagram.

2.5. Data extraction

Data from the included studies were extracted using a standardized data extraction spreadsheet developed for this review. For each study, information was systematically recorded regarding the authors and year of publication, the country or geographical context in which the research was conducted, sample number, the category of food or beverage investigated, and the type of nutrition labeling system tested. Additional information included the psychological mechanism evaluated (cognitive dissonance, psychological reactance, or both), measurement type (direct or indirect), the measurement methods used to assess psychological responses, and the main behavioral or attitudinal outcomes reported. The extracted information was subsequently reviewed to ensure accuracy and consistency across studies prior to the synthesis of results.

2.6. Methodological quality assessment

A methodological quality assessment of the included studies was performed using the Mixed Methods Assessment Tool (MMAT), version 2018 (Hong et al., 2018). This tool was selected because it allows for the assessment of heterogeneous study designs, including qualitative, descriptive quantitative, randomized controlled, and non-randomized studies. Each study was evaluated according to the criteria corresponding to its design category. The results of the assessment are presented in Supplementary Table S2 and were used to identify potential methodological limitations and sources of bias in the included studies.

2.7. Data synthesis

Given the heterogeneity of study designs, psychological constructs, labeling formats, food categories, and outcome measures, a quantitative meta-analysis was not considered appropriate. Therefore, the results were synthesized using a structured narrative approach. First, the studies were grouped according to the main psychological mechanism investigated: cognitive dissonance, psychological reactance, or both, and then compared according to the type of FOPL system examined, including warning labels, interpretive summary systems, and nutrient-specific formats.

Given the diversity of outcomes assessed across studies, findings were interpreted according to the different forms of consumer response reported in the literature, including emotional reactions, perceived threats to autonomy, risk perception, purchase intentions, actual food choices, information avoidance, and post-choice rationalization.

3. Results and discussion

3.1. Methodological quality assessment

The methodological quality assessment revealed that most studies met the majority of the MMAT criteria. However, several quantitative studies provided insufficient information regarding randomization procedures, baseline comparability, or blinding, resulting in “Can't tell” ratings for some criteria. In addition, many studies relied on hypothetical purchase scenarios, simulated food choices, or self-reported intentions rather than observed consumer behavior. The qualitative studies generally met all MMAT criteria, although their findings are limited in terms of generalizability. Detailed results of the quality assessment are presented in Supplementary Table S1.

3.2. Study selection and general characteristics of the included studies

The systematic search conducted across the selected databases initially identified 830 records, including 355 from PubMed, 6 from ScienceDirect, 465 from Scopus, and 4 from Web of Science. After the removal of 159 duplicate records, 671 articles remained for title and abstract screening. At this stage, 628 records were excluded because they did not meet the predefined eligibility criteria. The remaining 43 studies were retrieved for full-text assessment. After detailed evaluation, 35 articles retrieved from database searches were excluded because they did not explicitly investigate cognitive dissonance or psychological reactance in the context of nutrition labeling. Consequently, eight studies identified through database searches met the eligibility criteria. An additional four studies were identified through citation tracking and handsearching procedures, resulting in a final sample of 12 studies included in the review. The complete study selection process is presented in the PRISMA flow diagram (Fig. 1).

The studies included in this review were published between 2020 and 2025 and were conducted predominantly in high-income countries. This geographical concentration likely reflects both the earlier adoption of front-of-pack labeling systems in these regions and the stronger research infrastructure dedicated to food policy evaluation. Most studies employed experimental or quasi-experimental designs, often using simulated purchasing environments or controlled exposure to labeling interventions in order to measure psychological and behavioral responses.

Several of the included studies focused on ultra-processed products, particularly sugary beverages, snacks, and desserts (Fig. 2). This distribution likely reflects the policy priorities of many FOPL systems, which typically target products high in sugar, fat, and sodium. In addition,

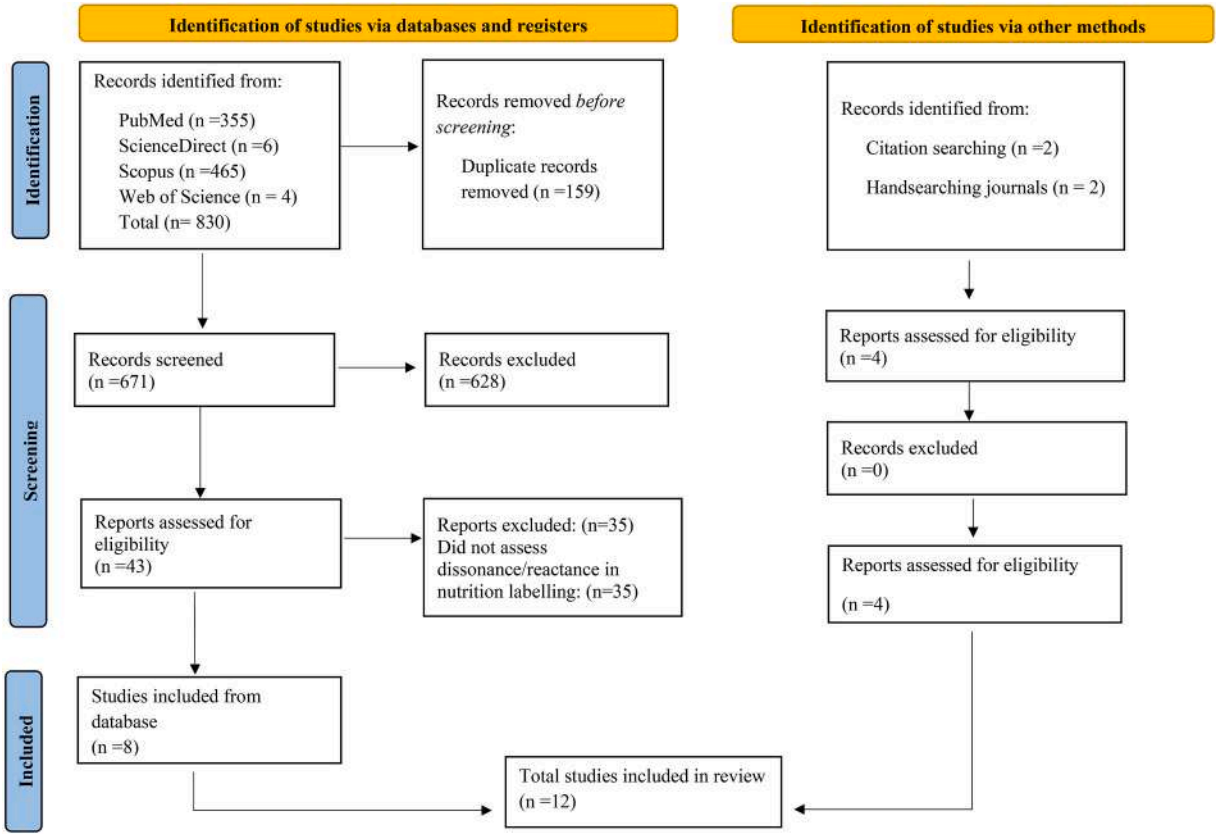


Fig. 1. PRISMA 2020 flow diagram for new systematic reviews which included searches of databases, registers and other sources.

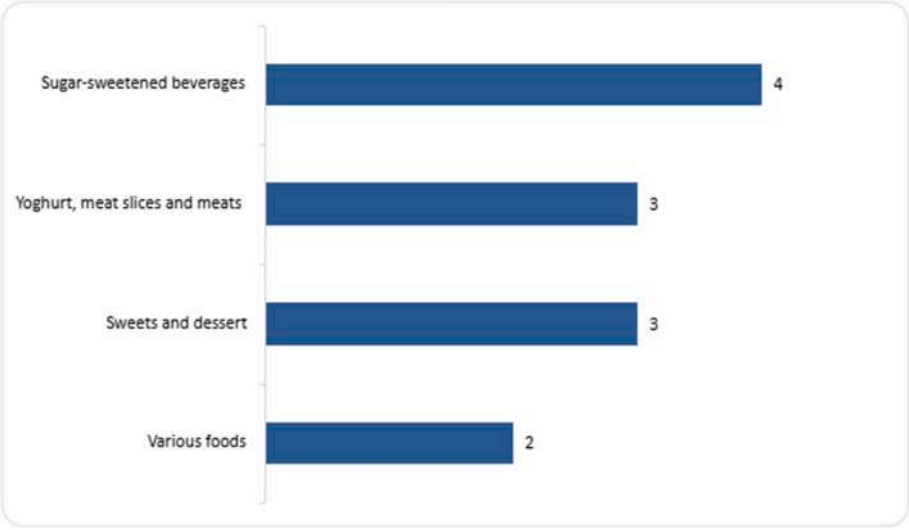


Fig. 2. Distribution of included studies according to food category.

these food categories often carry strong hedonic appeal, making them particularly relevant for investigating psychological resistance mechanisms. With respect to labeling systems, most included studies examined health warning labels - either textual or pictorial - and interpretive summary systems such as Nutri-Score.

Table 3 provides a synthesis of the included studies, summarizing the types of foods investigated, labeling formats, psychological mechanisms assessed, measurement approaches and key outcomes.

3.3. Operationalization of psychological mechanisms

Among the studies included in this review, psychological reactance was investigated more frequently than cognitive dissonance (Fig. 3). Moreover, substantial variability was observed in how these constructs were operationalized and measured.

Reactance was assessed using a variety of indicators, including perceived threats to freedom of choice, emotional responses such as anger or irritation, and behavioral indicators such as message rejection or defensive reactions. Some studies employed scales derived from the

Table 3
Synthesis of the included studies, psychological mechanisms assessed, measurement approaches, and main outcomes.

Author(s) and year	Country	Participants (N)	Food	Type of label	Mechanism assessed	Measurement	Measurement method	Main observed outcome
Clarke et al. (2020)	United Kingdom	4134	Various snacks	Health warning, calorie information	Reactance	Direct	7-point scale (avoidance, defensive reactions, acceptability)	↑ defensive reactions and ↓ acceptability, especially with images
Bullock et al. (2020)	USA	240	Ice cream	Nutrition table, ingredient list + health warning	Reactance	Indirect	Purchase choice	No behavioral change
Koch et al. (2022)	Various	472	Meat	Graphic image (cancer) + textual warning	Reactance	Direct	Brief Reaction to Health Warnings Scale	↑ Reactance
Miller, Wright, et al. (2022)	Australia	105	Sugar-sweetened beverages	Warnings (nutrients, exercise, pictograms)	Dissonance	Indirect	Qualitative analysis of responses	Text + pictograms ↑ dissonance
Miller, Dono, et al. (2022)	Australia	82	Sugar-sweetened beverages	Warnings (health effects, sugar content, exercise equivalents, pictograms)	Dissonance/ Reactance	Indirect	Qualitative analysis of responses	Text + pictogram (sugar spoons) → more dissonance; causal language → more rejection
Malik & Ong, 2024	Singapore	127	Soft drink	Health and pictorial warning	Reactance	Direct	EPPM scales + acceptability	Pictorial warnings evoked more reactance than textual ones
Mehlhose and Risius (2023)	Germany	1040	Candy and chocolate	Health warning + shocking images + "stop" sign	Reactance	Direct	EPPM +,choice experiment	Text generated reactance; images reduced the effect
Weingarten and Lagerkvist (2023)	Various	379	Meat	Health warning + images	Dissonance	Direct	7-point bipolar semantic scale	Text and images increased dissonance similarly
Schruff-Lim et al. (2024)	Netherlands	428	Yoghurt, meat slices, and meat products	Nutri-Score	Reactance	Indirect	7-point Likert (threat to freedom)	No observed reactance
Hémar-Nicolas et al. (2024)	France	15	Various	Nutri-Score/ Organic seal	Dissonance	Indirect	Semi-structured interviews	Cognitive confusion
Hughner & Dumitrescu, 2024	USA	190	Soft drink	Health warning	Reactance	Direct	5-point emotional scale	Significant reactance only with compliance cost
Yamim and Werle (2025)	France	147	Brownie	Nutri-Score	Dissonance	Indirect	7-point scale (guilt)	No direct change; Boomerang effect conditioned by expectations

Legend: EPPM = Extended Parallel Process Model.

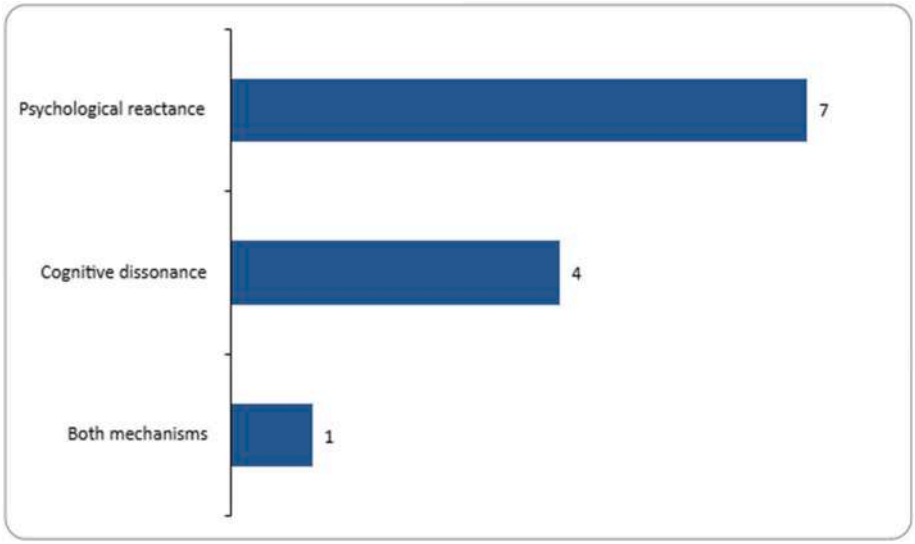


Fig. 3. Distribution of included studies according to psychological mechanism.

Extended Parallel Process Model (EPPM) or other validated instruments designed to capture emotional and motivational responses to persuasive messages. In contrast, cognitive dissonance was often measured through indicators such as feelings of guilt, psychological discomfort, self-reported conflict between beliefs and behaviour, or qualitative expressions of rationalization and justification.

This heterogeneity in measurement approaches complicates direct comparisons across studies and may partially explain the divergent

findings reported in the literature. The absence of standardized instruments for assessing psychological resistance in the context of food labeling highlights an important methodological challenge and underscores the need for greater conceptual and measurement consistency in future research. In addition to differences in the operationalization of cognitive dissonance and psychological reactance, the reviewed studies examined a wide range of consumer response outcomes. These included emotional responses (e.g., guilt, anger, irritation, and disgust), perceived threats to autonomy, changes in risk perception, purchase intentions, food choice outcomes, information avoidance, and post-choice rationalization. Distinguishing these outcome domains is important because they represent different manifestations of consumer response to nutrition labeling and should not be interpreted as equivalent forms of resistance. Throughout the following sections, these outcomes are discussed separately whenever relevant.

3.4. Psychological reactance: empirical divergences and moderating conditions

The empirical evidence regarding the emergence of psychological reactance in response to nutrition labeling interventions is not uniform. In particular, studies examining pictorial warning labels have produced divergent findings regarding their psychological and behavioral effects.

Clarke et al. (2020) and Malik and Ong (2024) observed that pictorial warnings elicited stronger reactance compared with text-only messages. Interestingly, these defensive reactions were accompanied by reductions in consumption intentions, suggesting that reactance does not necessarily prevent consumers from reconsidering their choices. In these cases, the emotional salience of visual stimuli may simultaneously increase perceived risk, producing a net reduction in consumption despite the presence of defensive responses. This finding illustrates that emotional reactance and behavioral intentions should not be interpreted as equivalent outcomes, as negative emotional responses may coexist with increased risk awareness and reduced consumption intentions.

In contrast, Mehlhose and Risius (2023) reported a different pattern. In their study, warning labels including graphic images reduced the likelihood of choosing sweets such as candies and chocolates while generating comparatively lower levels of reactance. The authors suggested that images may enhance perceived threat and may be associated with danger-control processes, which may increase message acceptance in some contexts.

Other design features also appear to influence reactance responses. For example, warning labels incorporating abstract symbols such as a "stop" sign showed weaker behavioral effects and, in some cases, were associated with increased product choice probabilities (Mehlhose & Risius, 2023). This finding suggests that symbols lacking strong emotional salience may be less effective in communicating risk and, in some contexts, may be associated with defensive responses.

The framing of warning messages also plays a crucial role. Miller, Dono, et al. (2022) observed that labels using causal language (e.g., "causes obesity") generated stronger reactance among parents of school-aged children, as these messages were perceived as authoritarian or judgmental. Conversely, informational messages expressing sugar content in teaspoons were perceived as more factual and less controlling, thereby reducing defensive reactions.

Similarly, Koch et al. (2022) found that graphic warnings linking meat consumption to cancer increased feelings of disgust and reduced consumption intentions but also triggered psychological reactance. Interestingly, the authors demonstrated that presenting the same information in a less explicitly persuasive format reduced defensive responses while maintaining behavioral impact.

Finally, contextual factors such as economic conditions may moderate reactance responses. Hughner and Dumitrescu (2024) showed that reactance toward sugar-sweetened beverage warnings increased when the cost of complying with the message was high. When pricing strategies encouraged larger portion sizes, consumers perceived greater economic barriers to behavioral change and reacted more defensively to warning labels. Conversely, when the cost of compliance was lower, reactance responses were attenuated.

Together, these findings suggest that psychological reactance does not operate as a simple barrier to behavioral change but rather as a context-dependent response shaped by message framing, emotional intensity, and perceived costs of compliance.

3.5. Cognitive dissonance: productive conflict or rationalization?

Compared with reactance, cognitive dissonance has been examined less frequently in the context of nutrition labeling. Nevertheless, the studies included in this review indicate that labeling interventions can activate cognitive conflict when presented information challenges existing beliefs, habits, or identities related to food consumption.

Miller, Dono, et al. (2022) found that parents exposed to warning labels on sugary beverages often experienced psychological discomfort when the message conflicted with their self-perception as responsible caregivers. In response, participants frequently employed rationalization strategies aimed at preserving their existing consumption behaviors. These strategies included questioning the causal claims of the label, emphasizing individual differences ("this does not apply to my child"), or invoking compensatory behaviors such as physical activity.

Similar patterns were observed among young adults in the study conducted by Miller, Wright, et al. (2022). Labels presenting sugar content in teaspoons generated strong emotional reactions and triggered cognitive conflict between existing consumption habits and newly salient nutritional information. In many cases, this dissonance was managed through self-exclusion strategies, whereby participants minimized the personal relevance of the information or reinterpreted the risks in ways that allowed them to maintain their behavior.

In other contexts, however, cognitive dissonance appeared to facilitate behavioral change. Weingarten and Lagerkvist (2023) demonstrated that labels highlighting the connection between animals and meat products increased dissonance among meat consumers and significantly increased the likelihood of choosing meat-free options.

Dissonance can also emerge when labeling information contradicts consumers' expectations regarding product healthfulness. Hémar-Nicolas et al. (2024) reported that Nutri-Score classifications triggered strong cognitive conflict when foods traditionally perceived as healthy received low ratings or when less healthy products obtained relatively favorable scores. This discrepancy generated confusion and sometimes led participants to question the credibility of the labeling system.

Similarly, Yamim and Werle (2025) showed that discrepancies between expected and actual Nutri-Score ratings generated feelings of guilt, which mediated changes in purchase intentions. In these cases, dissonance functioned as a motivational mechanism prompting consumers to reconsider their choices.

Overall, the evidence suggests that cognitive dissonance may operate as a double-edged mechanism: it can stimulate reflection and behavioral change, but it can also be resolved through rationalization or information avoidance.

3.6. Differences between warning labels and interpretive labeling systems

The studies included in this review primarily investigated two broad categories of front-of-pack labeling systems: warning-based labels and interpretive summary indicators such as Nutri-Score.

Warning labels, particularly those incorporating graphic imagery, tend to operate through the activation of strong emotional responses such as fear, disgust, or discomfort (Clarke et al., 2020; Hughner & Dumitrescu, 2024; Koch et al., 2022; Miller, Dono, et al., 2022; Yamim & Werle, 2025). These emotional reactions can increase perceived risk but may also trigger psychological reactance when consumers perceive the message as overly coercive or moralizing.

In contrast, interpretive labeling systems such as Nutri-Score primarily rely on cognitive processing mechanisms by providing simplified nutritional evaluations that guide consumers' decision-making. These systems often evoke emotions associated with moral evaluation or self-reflection, such as guilt or cognitive conflict, rather than the intense emotional responses typically associated with warning labels. Although confusion regarding the underlying scoring algorithm may occur in some cases (Hémar-Nicolas et al., 2024), interpretive systems are generally perceived as easier to understand and less intrusive.

Taken together, these findings suggest that different labeling systems may activate distinct psychological pathways, highlighting the importance of considering both emotional and cognitive mechanisms when designing nutrition labeling policies.

3.7. Food category effects

The evidence reviewed suggests that psychological responses to nutrition labeling vary across food categories. Nutritional variation across food categories. Animal-based products, particularly meat, may be associated with both cognitive dissonance and psychological reactance, given the strong cultural and identity-related meanings often attributed to these foods. Labels linking meat consumption to animal welfare or health risks can disrupt the psychological dissociation between animals and meat products, thereby generating cognitive conflict and increasing the likelihood of meat avoidance (Weingarten & Lagerkvist, 2023). At the same time, highly graphic warnings may trigger reactance if consumers perceive the message as manipulative or morally coercive (Koch et al., 2022).

Sugary beverages represent another category in which contextual factors strongly influence consumer responses. In these cases, reactance appears to be moderated not only by message framing but also by economic conditions such as pricing strategies and perceived costs of compliance (Hughner et al., 2024). Informational labels presenting sugar content in concrete units, such as teaspoons, tend to generate less reactance because they are perceived as factual rather than prescriptive (Miller, Dono, et al., 2022).

In indulgent foods, such as desserts and sweets, cognitive dissonance frequently arose when labeling contradicted consumers' expectations about the product's health. Worse-than-expected Nutri-Score ratings can generate feelings of guilt and reduce purchase intentions, consistent with expectation-disconfirmation processes (Yamim & Werle, 2025). On the other hand, textual warnings can sometimes be ignored in favor of sensory pleasure, especially in highly hedonic products such as ice cream (Bullock et al., 2020).

Finally, dairy products represented a context in which cognitive dissonance frequently arose due to conflicting beliefs about nutritional value. Products perceived as naturally healthy, such as yogurt or cheese, may generate confusion or resistance when assigned lower Nutri-Score ratings, leading consumers to question the labeling system or seek additional information to reconcile the inconsistency (Hémar-Nicolas et al., 2024).

Across different label formats, food categories, and study contexts, the reviewed evidence suggests that cognitive dissonance and psychological reactance should not be viewed as mutually exclusive responses to front-of-pack nutrition labels. Although the studies generally examined these mechanisms separately, some findings indicate that consumers may simultaneously experience conflict between existing beliefs and nutritional information while also perceiving labeling messages as threatening their freedom of choice. This coexistence may help explain why defensive emotional reactions do not always translate into rejection of health information or resistance to behavioral change. Rather, consumer responses appear to depend on the interaction between message characteristics, food category, prior beliefs, and perceived behavioral costs, resulting in a range of outcomes that include risk awareness, behavioral reconsideration, rationalization, information avoidance, or message rejection.

The relevance of these findings extends beyond consumer psychology. Front-of-pack nutrition labeling is increasingly implemented as part of broader public health strategies aimed at improving dietary patterns and reducing obesity-related health risks. The implications of these psychological responses extend beyond consumer decision-making and have potential relevance for broader nutrition and public health outcomes. Recent evidence highlights that obesity is a heterogeneous metabolic condition influenced by complex interactions among dietary exposures, inflammatory processes, insulin sensitivity, lipid metabolism, and other biological factors (Di Maio et al., 2026). In this context, understanding the psychological mechanisms that facilitate or hinder consumer engagement with nutrition labels becomes particularly important. Even the most scientifically accurate nutritional information may have limited public health impact if consumers respond defensively to the message or fail to integrate it into food-related decision-making processes. Consequently, psychological responses such as cognitive dissonance and reactance may represent important, although often overlooked, factors influencing the effectiveness of nutrition-related policies aimed at promoting healthier dietary behaviors and reducing cardiometabolic risk.

4. Limitations and future research

Some limitations should be considered when interpreting the findings of this review. First, the number of studies included in the final synthesis was relatively small (n = 12), reflecting the still limited body of research explicitly examining cognitive dissonance and psychological reactance in the context of nutrition labeling. This outcome is partly explained by the stringent inclusion criteria adopted in this review, which required studies to directly measure or explicitly interpret psychological resistance mechanisms. While this approach ensured conceptual clarity and analytical focus, it may have excluded studies that indirectly addressed similar processes without explicitly using these theoretical constructs.

Another limitation relates to the considerable methodological heterogeneity observed across the included studies. Differences in research designs, measurement approaches, and outcome variables were particularly evident in the operationalization of psychological constructs such as reactance and dissonance. This variability limits the comparability of results across studies and prevented the application of quantitative synthesis methods such as meta-analysis.

The geographical distribution of the evidence also represents an important limitation. Most studies were conducted in high-income countries, where front-of-pack labeling policies have been implemented earlier and where research infrastructures are more developed. Consequently, the extent to which these findings can be generalized to middle- and low-income contexts remains uncertain, especially given potential differences in food environments, dietary patterns, and levels of nutritional literacy. Furthermore, the existing literature shows a strong concentration on ultra-processed food categories, particularly sugary beverages, snacks, and desserts. Although these products represent key targets of nutrition labeling policies, this concentration limits the understanding of how psychological responses to labeling may operate in other food contexts.

The findings of this review should also be interpreted with caution, given the diversity of the available evidence. While some studies employed experimental designs assessing food choice outcomes or behavioral responses, others relied on self-reported intentions, perceptions, or qualitative accounts. Consequently, not all findings provide the same level of evidence regarding actual consumer behaviour. In particular, outcomes based on intentions or subjective perceptions should not be interpreted as direct indicators of behavioral change. Future research would benefit from greater use of real-world purchasing data and longitudinal designs to better understand how psychological responses to front-of-pack labels translate into sustained behavioral outcomes.

These limitations also point to several important directions for future research. First, greater theoretical integration between cognitive dissonance and psychological reactance is needed, as few studies have examined how these mechanisms interact or coexist in response to labeling interventions. Future research should also prioritize the development of more standardized measurement approaches for assessing psychological resistance mechanisms, which would facilitate cross-study comparisons and support cumulative knowledge development in this field. In addition, further investigation is needed regarding the moderating role of factors such as food identity, hedonic attachment to products, nutritional literacy, socioeconomic conditions, and perceived costs of behavioral change. Expanding research into real-world contexts and longitudinal designs would also provide valuable insights into how psychological responses to labeling evolve over time and influence actual purchasing behaviour. Finally, evaluating consumer responses to recently implemented labeling systems - such as the Brazilian front-of-pack "magnifying glass" model introduced by RDC ANVISA N° 429/2020 - may contribute to a better understanding of how different policy designs interact with cultural and market conditions.

By synthesizing empirical evidence through the lenses of cognitive dissonance and psychological reactance, this review contributes to a more nuanced understanding of the psychological mechanisms underlying consumer responses to front-of-pack nutrition labeling and helps explain why labeling policies may produce heterogeneous behavioral outcomes.

5. Conclusions

This systematic review demonstrates that front-of-pack nutrition labeling should not be understood as a purely informational intervention, but rather as a psychologically complex stimulus capable of eliciting different consumer responses. The reviewed evidence suggests that cognitive dissonance and psychological reactance are important mechanisms underlying heterogeneous reactions to nutrition labels, influenced by factors such as label format, food category, message framing, and consumers' pre-existing beliefs and motivations. Importantly, defensive responses do not necessarily preclude behavioral change. Across several studies, emotional resistance coexisted with increased risk perception, greater health awareness, and healthier intended choices, highlighting the complexity of consumer responses to front-of-pack labeling.

By integrating insights from cognitive dissonance and psychological reactance theories, this review contributes to a more comprehensive understanding of the psychological processes that may shape the effectiveness of front-of-pack nutrition labeling. The findings suggest that consumer resistance should not be interpreted solely as a barrier to intervention success, but rather as part of a broader process through which individuals interpret, negotiate, and respond to nutrition-related information.

Future research should prioritize the development of standardized measures of cognitive dissonance and psychological reactance in nutrition-labeling contexts, as well as the use of larger and more diverse samples, longitudinal designs, and real-world purchasing and consumption data. From a policy perspective, the findings indicate that nutrition-labeling systems may benefit from approaches that support consumer autonomy, communicate nutritional information transparently, avoid overly moralizing messages, and consider food category characteristics and consumer expectations. Such strategies may help reduce defensive responses while preserving the effectiveness of front-of-pack labeling as a public health tool.

CRediT authorship statement

Carolina de Souto Portel: Conceptualization, Methodology, Investigation, Data curation, Formal analysis, Writing - Original Draft, Visualization, Project administration.

Júlia B. Sousa: Methodology, Investigation, Data curation, Writing - Review & Editing.

Carolina P. C. Martins: Investigation, Validation, Writing - Review & Editing.

Mônica M. Pagani: Formal analysis, Writing - Review & Editing.

Eliane T. Mársico: Supervision, Data curation, Writing - Review & Editing.

Mônica Q. Freitas: Investigation, Supervision, Writing - Review & Editing.

Adriano G. Cruz: Conceptualization, Writing - Review & Editing.

Erick A. Esmerino: Conceptualization, Methodology, Supervision, Funding acquisition, Writing - Review & Editing, Project administration.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT in order to improve the language, grammar, and fluency of the manuscript. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tifs.2026.105892.

Data availability

Data will be made available on request.

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